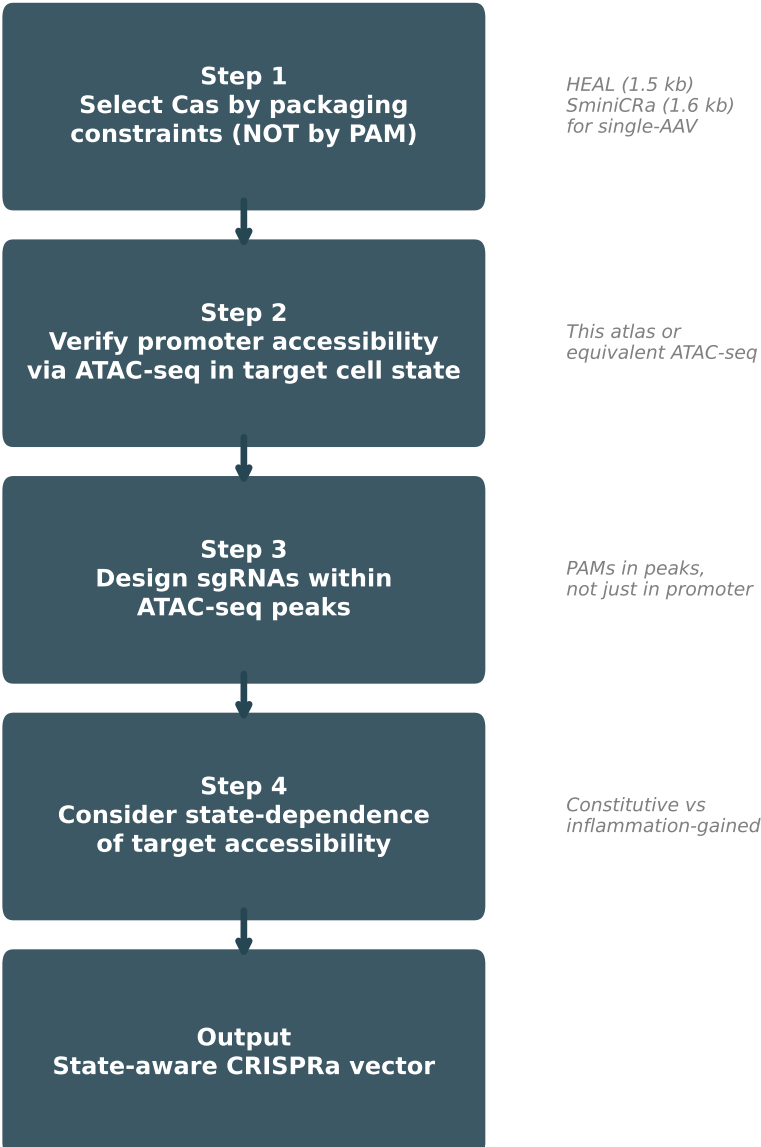


A Decision framework for in vivo CRISPRa design



B TFE3 sgRNA candidates: HEAL (TTTR) system
Constitutively accessible across all microglial states

#	Protospacer (5' to	Strand	GC%	Chromatin
1	GTTGAGCCACGCCTTCATT	—	50%	Open (all states)
2	CGTAGCTTCCTGAGCCACA	—	60%	Open (all states)
3	GCGGAACCATGCCGGGCTCT	—	65%	Open (all states)

*HEAL system: dUn1Cas12f1 + activation domain in single AAV
All candidates in ATAC-seq peaks across all 6 microglial states (3 labs)
Experimental validation required (heuristic pre-selection, not ML-predicted efficacy)*